

OptiFlow Allocation Engine Logic

This document provides a step-by-step description of the execution pipeline of the automated OptiFlow Allocation Engine.

Step 1: Initialization

- 1. Store Sorting:** Sorts scheduled stores (A, B, or C) and sub-sorts them by the custom priority list provided by the Center for Sight team.
 - 2. Deficit Calculation:** Calculates target displays (facings $\times$ depth) minus Current Stock on Hand (SOH).
 - **If Deficit = 0:** Stops execution for this store.
 - **If Deficit > 0:** Enters Tier 1 search loop.
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Step 2: Tier 1 Execution (Recent Sales Match)

- 1. Fetch Candidates:** Gathers exact SKUs sold in this store within the last X days.
 - 2. Apply Constraints:** Checks the candidate SKU against display rules:
 - **Max 3 Colors/Model:** Rejects SKU if store displays already contain 3 different colors of the same frame model.
 - **Max 2 Units/SKU:** Rejects SKU if dispatch allocation for this SKU is already at 2 units.
 - **Min 85% Brand Uniqueness:** Rejects SKU if it causes brand diversity to fall below 85% of active display displays.
 - **How:** If any check fails (NO), the SKU is discarded and the engine loops back to fetch the next Tier 1 candidate. If all checks pass (YES), it checks warehouse stock.
 - 3. Verify Warehouse Stock:**
 - **How:** If out of stock (NO), Tier 1 is exited, and the engine falls forward to Tier 2. If in stock (YES), units are allocated to the deficit.
 - 4. Check Deficit Filled:**
 - **How:** If the deficit is fully resolved (YES), execution moves to FIFO Batching. If not (NO), the engine loops back to fetch the next Tier 1 candidate.
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Step 3: Tier 2 Execution (Historical Sales Match)

Activates only if Tier 1 did not fully resolve the deficit.

- 1. Fetch Candidates:** Gathers exact matching SKUs sold in this store in the last 6 months.
 - 2. Apply Constraints:** Checks the same rules (3 colors/model, 2 units/SKU, 85% brand uniqueness).
 - **How:** If failed (NO), discards SKU and loops back to fetch the next Tier 2 candidate. If passed (YES), checks warehouse stock.
 - 3. Verify Warehouse Stock:**
 - **How:** If out of stock (NO), Tier 2 is exited, and the engine falls forward to Tier 3. If in stock (YES), units are allocated.
 - 4. Check Deficit Filled:**
 - **How:** If deficit is fully resolved (YES), execution moves to FIFO Batching. If not (NO), the engine loops back to fetch the next Tier 2 candidate.
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Step 4: Tier 3 Execution (Substitute Cascade)

Activates only if Tier 1 and Tier 2 did not fully resolve the deficit.

- 1. Fetch Candidates:** Finds warehouse frames priced within $\pm 20\%$ of the original SKU, progressively relaxing matching filters in this order: Color $\rightarrow$ Material $\rightarrow$ Shape.
 - 2. Apply Constraints:** Checks the same rules (3 colors/model, 2 units/SKU, 85% brand uniqueness).
 - **How:** If failed (NO), discards substitute and loops back to fetch the next Tier 3 candidate. If passed (YES), checks warehouse stock.
 - 3. Verify Warehouse Stock:**
 - **How:** If out of stock (NO), allocation for this deficit fails (left unfilled). Execution moves to FIFO Batching. If in stock (YES), substitute units are allocated.
 - 4. Check Deficit Filled:**
 - **How:** If deficit is resolved (YES), execution moves to FIFO Batching. If not (NO), the engine loops back to find the next Tier 3 substitute candidate.
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Step 5: Final Batching & Dispatch

- 1. FIFO Batching:** If an allocated SKU has multiple batches in the warehouse, the engine selects the inventory batch with the oldest arrival timestamp first.

2. Dispatch: Generates the final pick-list order and exports it to the HITL dashboard.